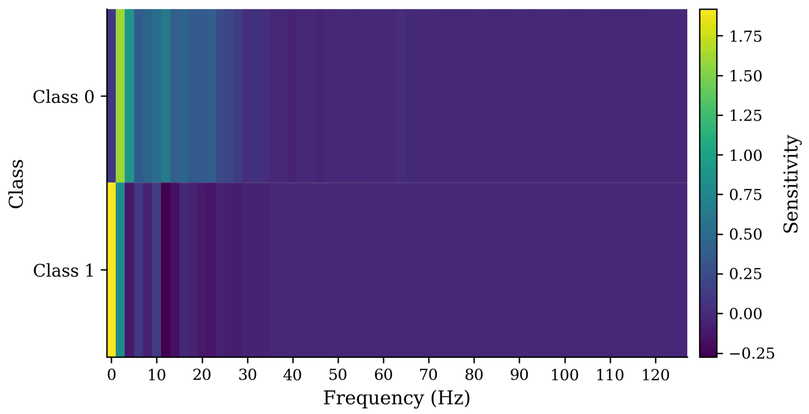
Sensitivity

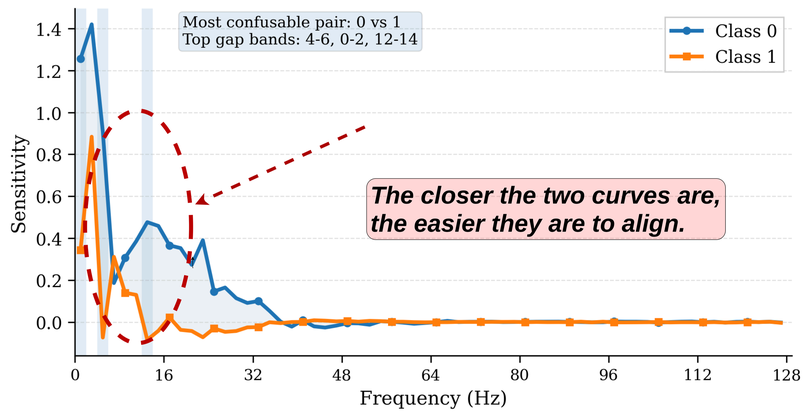
$$s_i(b) = f_{y_i}(x_i) - f_{y_i}(x_i^{-b})$$

Here $f_{y_i}(x_i)$ denotes the model score (logit or probability) for the true class of sample x_i, and x_i^{-b} represents the signal with the b-th frequency band removed. The sensitivity $s_i(b)$ measures how much the model's confidence in the correct class decreases when a specific frequency band is masked. A larger value indicates that the band is more important for correctly recognizing that class.

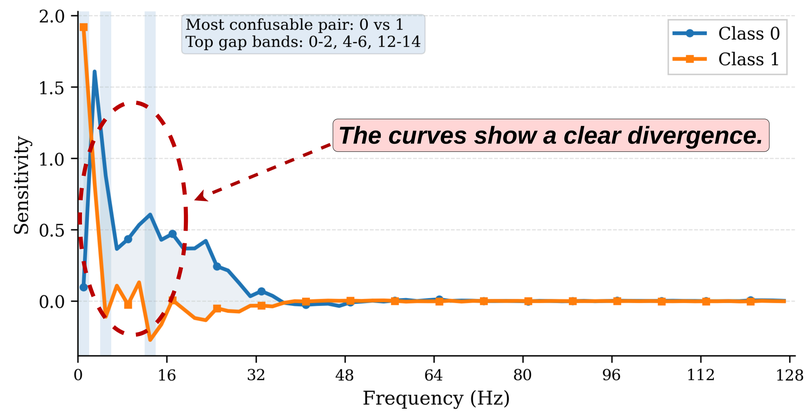
Result



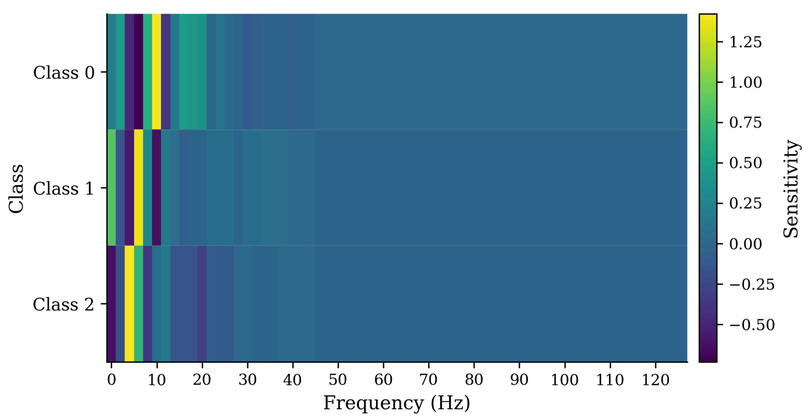
(a) PTB Heatmap



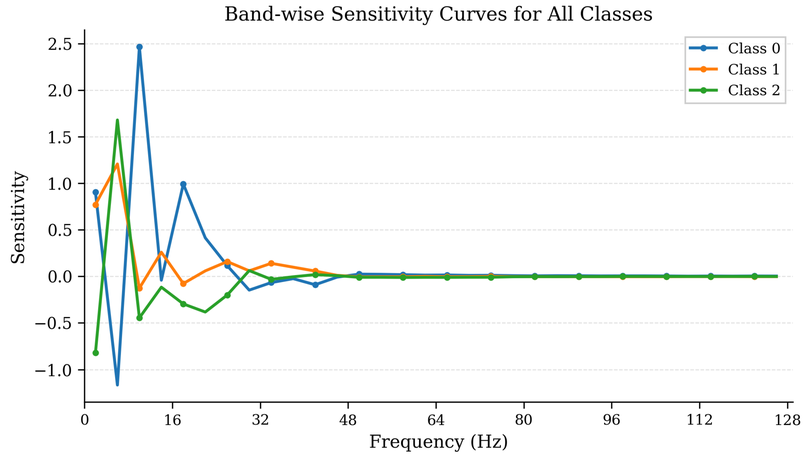
(b) PTB w/o SCLN



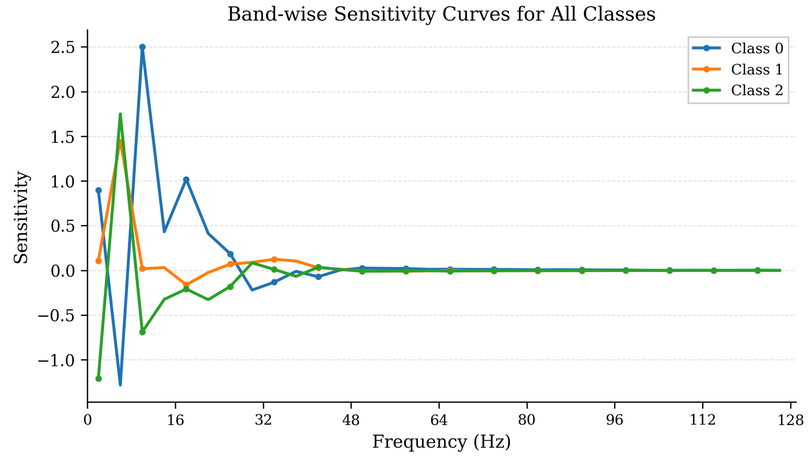
(c) PTB w/ SCLN



(d) ADFTD Heatmap



(e) ADFTD w/o SCLN



(f) ADFTD w/ SCLN

Analysis

The figure presents a band-wise sensitivity analysis on both the PTB and ADFTD datasets, including heatmaps and sensitivity curves. For the PTB dataset ((a), (b), and (c)), the heatmap (a) shows that discriminative information is mainly concentrated in low-frequency regions, while the corresponding curves ((b) and (c)) indicate that, without SCLN, the sensitivities of the two classes are relatively close in key bands, suggesting limited separability; after applying SCLN, the curves become more distinct, reflecting improved class-specific feature learning. Similarly, for the ADFTD dataset ((d), (e), and (f)), the heatmap (d) reveals a comparable low-frequency dominance, and the curves ((e) and (f)) demonstrate clearer divergence between classes with SCLN. Overall, the results consistently show that SCLN enhances the separation of class-dependent spectral features across both datasets.

Conclusion

Guided implicitly by labels, SCLN adaptively preserves different information depending on the data and modality.